38 Adaptive Communication in Cultural Heritage Conversational Agents: A Prompt-Based Approach for User Expertise-Aware Responses

Gioele Litrico (gioele.litrico@phd.unict.it), Giovanni Gallo, Filippo Stanco

Conversational agents are increasingly being adopted in museums and cultural heritage to provide visitors with interactive access to cultural information tailored needs. Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) enable natural language interaction while grounding responses in curated institutional sources, improving factual consistency. However, conventional agents generally adopt the same communication style regardless of visitors' knowledge or curiosity.

In this study, we propose a prompt-based approach that enables a conversational agent to infer the user's level of expertise and curiosity from their questions and adapt its linguistic style accordingly. We define three communication strategies: (i) concise, for users seeking brief and accessible information; (ii) exploratory, for non-expert users expressing greater curiosity and requesting deeper explanations; and (iii) technical, for users whose questions indicate higher expertise and require precise, domain-specific terminology and detailed context.

For each strategy, we developed dedicated prompting instructions and created a questionanswer dataset of 75 questions, 25 for each profile, derived from the catalogue of the Museum of Archaeology of the University of Catania (MAUC). We evaluated two LLMs, GPT-OSS20B and Gemma 4 E4B, comparing responses with and without the conditioning prompt. Five independent LLM judges assessed the correspondence between generated and reference answers on a 1–5 scale, focusing on communication strategy. The proposed prompt substantially improved performance, increasing the average score from approximately 3.5 to 4.5. GPT-OSS-20B achieved the best result (4.8/5), while Gemma 4 E4B obtained lower but still acceptable scores.

Overall, these results indicate that prompt-based conditioning can effectively guide LLMs toward user-adaptive communication, supporting more personalized conversational experiences in cultural heritage applications